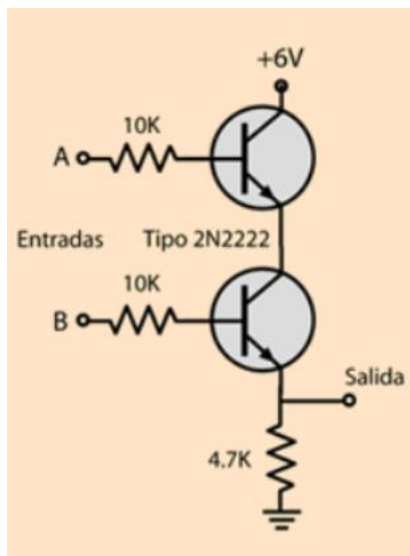


Compuertas Logicas

AND

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

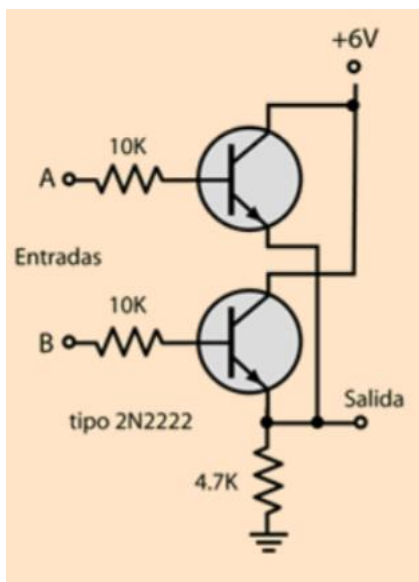


Sintaxis verilog: **&**: (assign y = a & b;)

Ó: **and**: (and u1 (y, a, b);)

OR

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

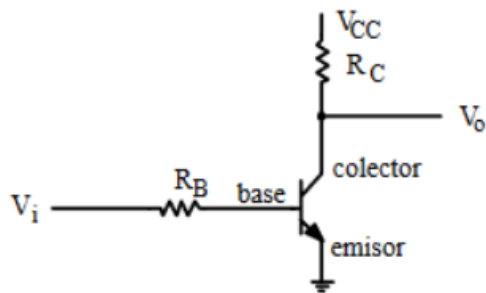
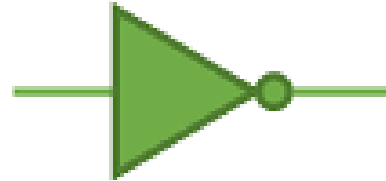


Sintaxis verilog: |: assign y = a | b;

Ó: or: or u1 (y, a, b);

NOT

Q	Q'
0	1
1	0



Inversor:

$V_i = 0 = "0"$ → corte → $V_o = V_{cc} = "1"$

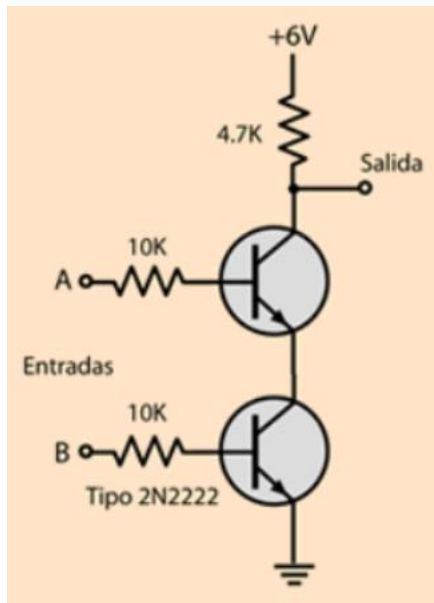
$V_i = V_{cc} = "1"$ → saturación → $V_o = 0 = "0"$

Sintaxis en verilog: `~`: assign $y = \sim a$; Ó **not**: not n1 (y, a);

NAND

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0



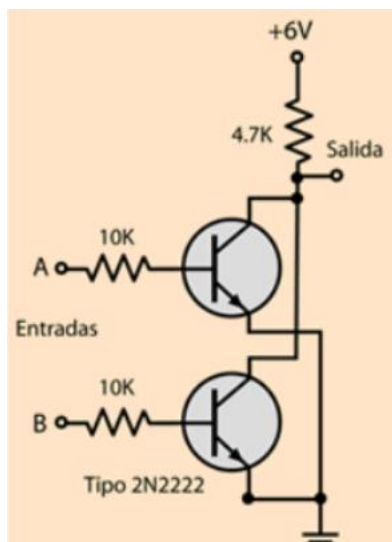


Sintaxis en verilog: **nand**: nand (Y, A, B);

Ó: assign Y = ~(A & B);

NOR

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

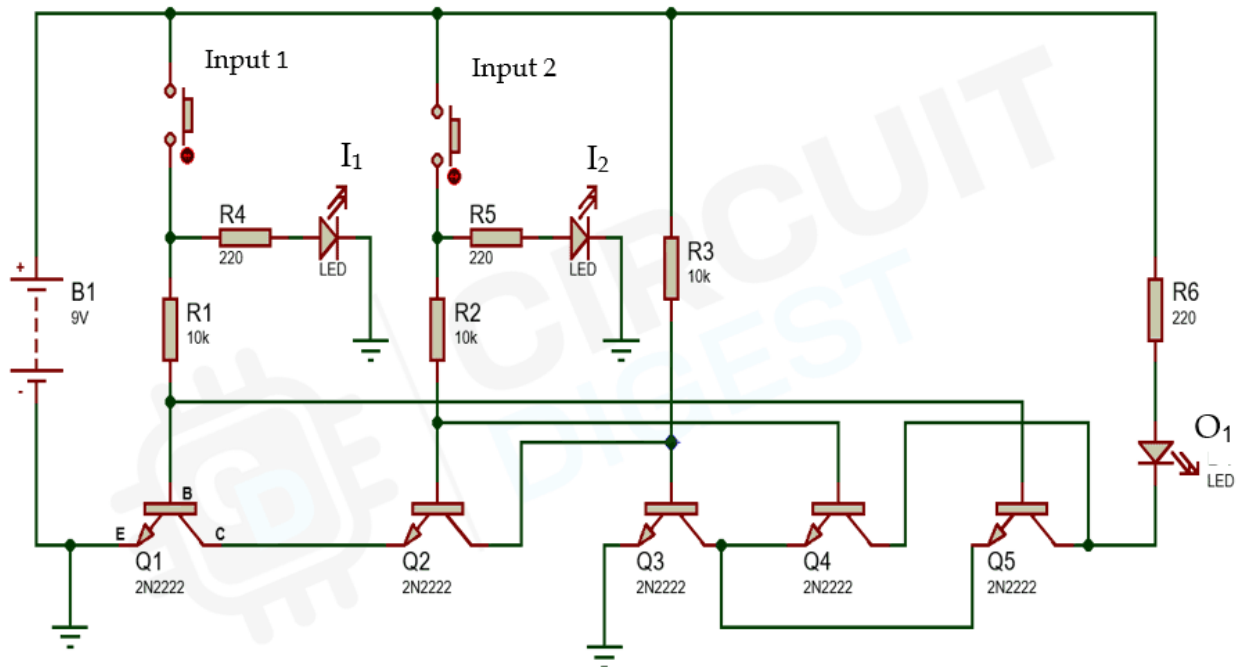


Sintaxis de verilog: assign y = ~(a | b);

Ó: nor (y, a, b);

XOR

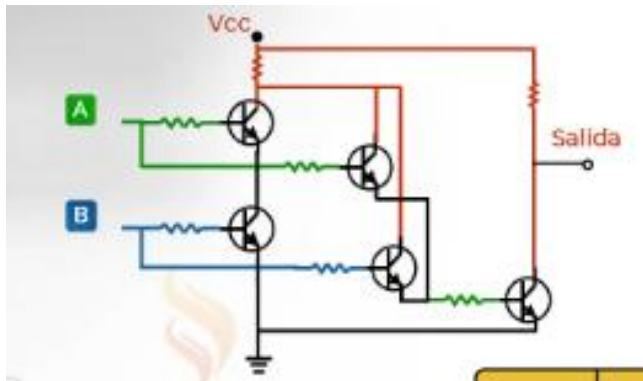
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0



Sintaxis en verilog: $\wedge$ assign $y = a \wedge b$; Ó : **xor** gate1 (y, a, b);

XNOR

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	1



Sintaxis en verilog: `assign y = ~(a ^ b);` Ó: `xnor inst_xnor (y, a, b);`